# Chapter 10. Analysis of Variance and Regression

One-way and two-way ANOVA, quadratic forms, linear regression

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## 10.1 One-way analysis of variance

The **one-way normal model** (Chapter 4) is

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad i = 1, \dots, k,\ j = 1, \dots, n_i,$$

models a single **factor** with $k$ **levels** (**treatments**). With $n = \sum n_i$, overall effect $\bar{\mu} = \sum_i n_i \mu_i / n$ and treatment effects $\alpha_i = \mu_i - \bar{\mu}$,

$$X_{ij} = \bar{\mu} + \alpha_i + e_{ij}, \qquad \sum_i n_i \alpha_i = 0.$$

**Theorem 10.1 (UMVUEs)**

In the one-way normal model, the UMVUEs are

$$\hat{\mu}_i = \bar{X}_i = \frac{1}{n_i} \sum_j X_{ij}, \qquad \hat{\bar{\mu}} = \sum_i \frac{n_i}{n} \hat{\mu}_i, \qquad \hat{\alpha}_i = \hat{\mu}_i - \hat{\bar{\mu}}, \qquad \hat{\sigma}^2 = \frac{1}{n-k} \sum_{i,j} (X_{ij} - \bar{X}_i)^2.$$

**Proof.** The joint pdf $\exp\{-\frac{1}{2\sigma^2}\sum_{i,j}(x_{ij}-\mu_i)^2 - \frac{n}{2}\log(2\pi\sigma^2)\}$ is a $(k+1)$-parameter exponential family meeting Theorem 8.8, so $(\bar{X}_1, \dots, \bar{X}_k, \sum_{i,j}(X_{ij} - \bar{X}_i)^2)$ is complete sufficient; the estimators are unbiased functions of it (Theorem 4.9 in Volume I), hence UMVUEs (Lehmann–Scheffé). ■

The MLE of $\sigma^2$ is $\sum (X_{ij} - \bar{X}_i)^2 / n$; the other MLEs are the UMVUEs.

**Theorem 10.2 (Sampling distributions)**

(a) $\hat{\mu}_i = \bar{X}_i$ are independent, $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$.
(b) $\hat{\sigma}^2$ is independent of $(\bar{X}_1, \dots, \bar{X}_k)$.
(c) $(n-k)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-k)$.

**Proof.** By Theorem 4.5 in Volume I per group, $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$ is independent of $SS_i = \sum_j (X_{ij} - \bar{X}_i)^2$, $SS_i/\sigma^2 \sim \chi^2(n_i - 1)$; groups are independent and $(n-k)\hat{\sigma}^2 = \sum_i SS_i$. ■

### Simultaneous confidence intervals

Let $\hat{\mu} = (\hat{\mu}_1, \dots, \hat{\mu}_k)'$, $D = \mathrm{Var}(\hat{\mu})/\sigma^2 = \mathrm{diag}(1/n_i)$, and for a rank-$r$ $k \times r$ matrix $C$, $\mathrm{col}(C) = \{C\mathbf{a} : \mathbf{a} \in \mathbb{R}^r\}$.

**Theorem 10.3 (Confidence sets and Scheffé intervals)**

(a) $P\{(C'\mu - C'\hat{\mu})'(C'DC)^{-1}(C'\mu - C'\hat{\mu}) \le r\hat{\sigma}^2 F_\alpha(r, n-k)\} = 1 - \alpha$.

(b) $P\Big\{|\mathbf{c}'\mu - \mathbf{c}'\hat{\mu}| \le \sqrt{\textstyle\sum_i c_i^2 \hat{\sigma}^2/n_i}\sqrt{r F_\alpha(r, n-k)} \ \text{ for all } \mathbf{c} \in \mathrm{col}(C)\Big\} = 1 - \alpha$.

**Proof.** (a) $C'\hat{\mu} \sim N(C'\mu, \sigma^2 C'DC)$, so the quadratic form over $\sigma^2$ is $\chi^2(r)$ (Theorem 4.18 in Volume I), independent of $(n-k)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-k)$; the ratio is $F(r, n-k)$.

(b) For $\mathbf{c} = C\mathbf{a}$, $\mathbf{d} = C'(\mu - \hat{\mu})$, $M = C'DC$, by Cauchy–Schwarz

$$\max_{\mathbf{c} \in \mathrm{col}(C), \mathbf{c} \neq \mathbf{0}} \frac{\{\mathbf{c}'(\mu - \hat{\mu})\}^2}{\mathbf{c}' D \mathbf{c}} = \max_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}'\mathbf{d})^2}{\mathbf{a}' M \mathbf{a}} = \max_{\mathbf{b} \neq \mathbf{0}} \frac{(\mathbf{b}' M^{-1/2} \mathbf{d})^2}{\mathbf{b}'\mathbf{b}} = \mathbf{d}' M^{-1} \mathbf{d}.$$

so (b) and (a) are the same event. ■

$\sum c_i\mu_i$ with $\sum c_i = 0$ is a **contrast**, e.g. $\mu_i - \mu_l = \alpha_i - \alpha_l$; contrasts form a $(k-1)$-dimensional space:

**Theorem 10.4 (Simultaneous intervals for contrasts)**

(a) (Scheffé) With probability $1-\alpha$, for all $\mathbf{c}$ with $\sum c_i = 0$,

$$|\mathbf{c}'\alpha - \mathbf{c}'\hat{\alpha}| \le \sqrt{\textstyle\sum_i c_i^2\hat{\sigma}^2/n_i}\,\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(b) In particular, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(c) (Bonferroni) With $m = k(k-1)/2$ and $\alpha^* = \alpha/m$, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\; t_{\alpha^*/2}(n-k).$$

**Proof.** (a) Theorem 10.3(b) on the contrast space;

(b) is a sub-event.

(c) Per pair $r=1$ and $F_{\alpha^*}(1,n-k) = t_{\alpha^*/2}(n-k)^2$, coverage $1-\alpha^*$; by Bonferroni $P(A_1\cap\cdots\cap A_m) \ge 1 - m\alpha^* = 1-\alpha$. ■

Bonferroni is shorter for few comparisons; Scheffé covers all contrasts. E.g. $n=20$, $\alpha=0.05$: $k=3$ gives multipliers 2.680 (Scheffé) vs 2.654 (Bonferroni); $k=7$, 4.182 vs 3.760.

### Testing equality of treatments

**Theorem 10.5 ($F$ test)**

With $SSB = \sum_i n_i(\bar{X}_i - \bar{X})^2$ (between) and $SSW = \sum_{i,j}(X_{ij}-\bar{X}_i)^2 = (n-k)\hat{\sigma}^2$ (within), the LRT of $H_0 : \mu_1 = \cdots = \mu_k$ rejects when

$$F_n = \frac{SSB/(k-1)}{SSW/(n-k)} \ge F_\alpha(k-1,n-k).$$

**Proof.** $\hat{\theta}^{\Omega} = (\bar{x}_1,\ldots,\bar{x}_k,\hat{\sigma}^{2,\Omega})$, $n\hat{\sigma}^{2,\Omega} = SSW$; $\hat{\theta}^{\Omega_0} = (\bar{x},\ldots,\bar{x},\hat{\sigma}^{2,0})$, $n\hat{\sigma}^{2,0} = SSW + SSB$. So

$$2\{l(\hat{\theta}^{\Omega}) - l(\hat{\theta}^{\Omega_0})\} = n\log\frac{\hat{\sigma}^{2,0}}{\hat{\sigma}^{2,\Omega}} = n\log\Big(1+\frac{SSB}{SSW}\Big),$$

increasing in $F_n$. The map $A : \mathbf{X} \mapsto (\bar{X}_i - \bar{X})$ is symmetric idempotent with $SSB = \mathbf{X}'A\mathbf{X}$, $\mathrm{tr}A = \sum_{i,j}(1/n_i - 1/n) = k-1$, $A\mathbf{1} = \mathbf{0}$; under $H_0$, $SSB/\sigma^2 = \mathbf{Z}'A\mathbf{Z} \sim \chi^2(k-1)$, $\mathbf{Z} = (\mathbf{X} - \bar{\mu}\mathbf{1})/\sigma$, independent of $SSW/\sigma^2 \sim \chi^2(n-k)$ (Theorem 10.2). ■

With unweighted effects $\tilde{\alpha}_i = \mu_i - \sum\mu_i/k$, the null is the same and contrasts agree: $\sum c_i\tilde{\alpha}_i = \sum c_i\alpha_i$.

**Example 10.1**

Yields (in kg) of a crop under three fertilizers:

| fertilizer | observations | $n_i$ | $\bar{x}_i$ |
|---|---|---|---|
| A | 20, 22, 19, 23 | 4 | 21 |
| B | 25, 27, 26 | 3 | 26 |
| C | 22, 21, 24, 23, 25 | 5 | 23 |

$n = 12$, $\bar{x} = 23.083$, $SSB = 42.92$, $SSW = 22$: $F = (42.92/2)/(22/9) = 8.78 > F_{0.05}(2,9) = 4.26$. Bonferroni ($m = 3$, $t_{0.05/6}(9) = 2.933$, $\hat{\sigma}^2 = 2.444$) for $\mu_B - \mu_A$: $5 \pm 2.933\sqrt{2.444(1/3+1/4)} = 5 \pm 3.50$, excluding 0.

## 10.2 Two-way analysis of variance

Factors $A$ ($i \le a$) and $B$ ($j \le b$); cell $(i,j)$ observed $n_{ij}$ times:

$$X_{ijk} = \mu_{ij} + e_{ijk}, \qquad e_{ijk} \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2), \qquad k = 1,\dots,n_{ij}.$$

A one-way model with $ab$ treatments, so:

**Theorem 10.6**

(a) The UMVUEs are $\hat{\mu}_{ij} = \bar{X}_{ij}$ and $\hat{\sigma}^2 = \sum_{i,j,k}(X_{ijk} - \bar{X}_{ij})^2/(n-ab)$, $n = \sum n_{ij}$.
(b) The $\bar{X}_{ij}$ are independent $N(\mu_{ij}, \sigma^2/n_{ij})$.
(c) $\hat{\sigma}^2$ is independent of the cell means.
(d) $(n-ab)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-ab)$.
(e) Confidence sets and Scheffé intervals are as in Theorem 10.3, with $D = \operatorname{diag}(1/n_{ij})$ and $n - ab$ degrees of freedom.

The effects are defined by

$$\bar{\mu}_{\cdot\cdot} = \sum_{i,j} \frac{n_{ij}}{n}\mu_{ij}, \quad \bar{\mu}_{i\cdot} = \sum_j \frac{n_{ij}}{n_{i\cdot}}\mu_{ij}, \quad \bar{\mu}_{\cdot j} = \sum_i \frac{n_{ij}}{n_{\cdot j}}\mu_{ij},$$

$$\alpha_i = \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}, \qquad \beta_j = \bar{\mu}_{\cdot j} - \bar{\mu}_{\cdot\cdot}, \qquad \gamma_{ij} = \mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}.$$

$\alpha_i, \beta_j$ are the **main effects**, $\gamma_{ij}$ (the non-additive part) the **interaction**. In the **balanced** case $n_{ij} \equiv r$,

$$X_{ijk} = \bar{\mu}_{\cdot\cdot} + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad \sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0.$$

**Theorem 10.7 (Pairwise comparisons of main effects)**

In the balanced model, $\hat{\alpha}_i = \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots}$, $n_{i\cdot} = rb$, $n = rab$, and

(a) with probability at least $1-\alpha$, for all $i \ne l$, $|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{2\hat{\sigma}^2/(rb)}\sqrt{(a-1)F_\alpha(a-1, n-ab)}$;
(b) with $m = a(a-1)/2$, $\alpha^* = \alpha/m$, the same holds with $t_{\alpha^*/2}(n-ab)$ in place of $\sqrt{(a-1)F_\alpha(a-1,n-ab)}$.

**Proof.** $\alpha_i - \alpha_l = \sum_j(\mu_{ij} - \mu_{lj})/b$ is a contrast with $\sum c^2/n_{ij} = 2/(rb)$, in an $(a-1)$-dimensional space; apply Theorem 10.6(e) and Theorem 10.4. ■

**Theorem 10.8 (Test for interaction)**

In the balanced model let

$$SS_{AB} = \sum_{i,j} r(\bar{X}_{ij\cdot} - \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdot j\cdot} + \bar{X}_{\cdots})^2, \qquad SSE = \sum_{i,j,k} (X_{ijk} - \bar{X}_{ij\cdot})^2.$$

The LRT of $H_0^{AB}: \gamma_{ij} = 0$ for all $i, j$ (the additive model) rejects when

$$F_n = \frac{SS_{AB}/\{(a-1)(b-1)\}}{SSE/(n-ab)} \geq F_\alpha((a-1)(b-1), n-ab).$$

**Proof.** Under $\Omega$, $\hat{\mu}_{ij} = \bar{x}_{ij\cdot}$, $n\hat{\sigma}^{2,\Omega} = SSE$; under $H_0^{AB}$ (balanced), $\hat{\mu}_{ij}^0 = \bar{x}_{i\cdot\cdot} + \bar{x}_{\cdot j\cdot} - \bar{x}_{\cdots}$ and by orthogonality $n\hat{\sigma}^{2,0} = SSE + SS_{AB}$, so the statistic is $n\log(1 + SS_{AB}/SSE)$. The map $\Pi: (x_{ijk}) \mapsto (\bar{x}_{ij\cdot} - \bar{x}_{i\cdot\cdot} - \bar{x}_{\cdot j\cdot} + \bar{x}_{\cdots})$ is symmetric idempotent, $SS_{AB} = \mathbf{X}'\Pi\mathbf{X}$, and

$$\mathrm{tr}\Pi = \sum_{i,j,k} \left(\frac{1}{r} - \frac{1}{rb} - \frac{1}{ra} + \frac{1}{rab}\right) = ab - a - b + 1 = (a-1)(b-1).$$

Under $H_0^{AB}$, $\Pi$ annihilates the mean, so $SS_{AB}/\sigma^2 \sim \chi^2((a-1)(b-1))$, a function of cell means, independent of $SSE/\sigma^2 \sim \chi^2(n-ab)$. ■

**Theorem 10.9 (Tests for main effects)**

In the balanced model, with $SS_A = \sum_i br(\bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots})^2$ and $SS_B = \sum_j ar(\bar{X}_{\cdot j\cdot} - \bar{X}_{\cdots})^2$:

(a) the LRT of $H_0^A: \alpha_1 = \cdots = \alpha_a = 0$ rejects when $\dfrac{SS_A/(a-1)}{SSE/(n-ab)} \geq F_\alpha(a-1, n-ab)$;

(b) the LRT of $H_0^B: \beta_1 = \cdots = \beta_b = 0$ rejects when $\dfrac{SS_B/(b-1)}{SSE/(n-ab)} \geq F_\alpha(b-1, n-ab)$.

Proofs as for Theorem 10.8. Summary ($\delta$: noncentrality, next section):

| source | SS | df | MS | $F$ | $\delta$ |
|---|---|---|---|---|---|
| $A$ | $SS_A$ | $a-1$ | $MS_A$ | $MS_A/MSE$ | $\sum br\alpha_i^2/\sigma^2$ |
| $B$ | $SS_B$ | $b-1$ | $MS_B$ | $MS_B/MSE$ | $\sum ar\beta_j^2/\sigma^2$ |
| $A \times B$ | $SS_{AB}$ | $(a-1)(b-1)$ | $MS_{AB}$ | $MS_{AB}/MSE$ | $\sum r\gamma_{ij}^2/\sigma^2$ |
| error | $SSE$ | $n-ab$ | $MSE$ | | |
| total | $SST$ | $n-1$ | | | |

## 10.3 Power of $F$ tests

**Motivation.** Under alternatives $F$ is noncentral, and the noncentrality determines power and sample size.

**Theorem 10.10 (Noncentral chi-square)**

Let $X_i \sim N(\mu_i, 1)$, $i = 1, \ldots, r$, be independent, $Y = X_1^2 + \cdots + X_r^2$ and $\delta = \mu_1^2 + \cdots + \mu_r^2$. Then

(a) $K_Y(t) = -\frac{r}{2}\log(1-2t) + \frac{\delta}{2} \cdot \frac{2t}{1-2t} = \sum_{k\geq 1} \frac{t^k}{k!} 2^{k-1}(k-1)!\,(r + k\delta)$ for $t < 1/2$;

(b) with $p_{P(\delta/2)}(k)$ the Poisson$(\delta/2)$ pmf, $f_Y(y) = \sum_{k\ge 0} p_{P(\delta/2)}(k)\, f_{\chi^2(r+2k)}(y)$.

So the law depends only on $\delta$: the **noncentral chi-square** $\chi^2(r;\delta)$.

**Proof.** (a) For $X \sim N(\mu,1)$, $t < 1/2$, completing the square,

$$Ee^{tX^2} = \int \frac{1}{\sqrt{2\pi}} \exp\Big\{-\frac{1}{2}(1-2t)\Big(x - \frac{\mu}{1-2t}\Big)^2 + \frac{t\mu^2}{1-2t}\Big\}dx = (1-2t)^{-1/2}\exp\Big(\frac{\mu^2 t}{1-2t}\Big).$$

Multiply over $i$, take logs, and expand $-\log(1-2t) = \sum (2t)^k/k$, $\frac{2t}{1-2t} = \sum_{k\ge 1}(2t)^k$.

(b) By (a), $Y \overset{d}{=} V + Z^2$, $V \sim \chi^2(r-1)$ independent of $Z \sim N(\sqrt{\delta},1)$. With $\theta = \sqrt{\delta}$,

$$f_{Z^2}(x) = \frac{1}{2\sqrt{x}}\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x - \frac{1}{2}\theta^2}\big(e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}}\big) = \sum_{k\ge 0} p_{P(\delta/2)}(k) f_{\chi^2(1+2k)}(x),$$

expanding $e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}} = 2\sum_k \theta^{2k}x^k/(2k)!$ and using $\Gamma(\tfrac{1}{2})\Gamma(2k+1) = 2^{2k}\Gamma(k+\tfrac{1}{2})\Gamma(k+1)$; convolve with $\chi^2(r-1)$. ■

**Theorem 10.11**

If $\mathbf{X} \sim N(\mu, I)$ and $A$ is symmetric idempotent, then $\mathbf{X}'A\mathbf{X} \sim \chi^2(r;\delta)$ with $r = \operatorname{tr} A$ and $\delta = \mu' A \mu$.

**Proof.** $A = P\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P'$, $P$ orthogonal; $\mathbf{Z} = P'\mathbf{X} \sim N(\eta, I)$, $\eta = P'\mu$, and $\mathbf{X}'A\mathbf{X} = Z_1^2 + \cdots + Z_r^2$ with $\eta_1^2 + \cdots + \eta_r^2 = \mu' A \mu$. ■

For independent $V_1 \sim \chi^2(r_1;\delta)$, $V_2 \sim \chi^2(r_2)$, $(V_1/r_1)/(V_2/r_2) \sim$ **noncentral** $F$ $F(r_1, r_2; \delta)$.

**Example 10.2 (Power of the one-way $F$ test)**

$SSB = \mathbf{X}'A\mathbf{X}$ with $A$ idempotent of trace $k-1$, mapping $(\mu_i)$ to $(\alpha_i)$; by Theorem 10.11

$$SSB/\sigma^2 \sim \chi^2(k-1;\delta), \qquad \delta = \sum_i n_i \alpha_i^2/\sigma^2,$$

so the power is $P\{F(k-1, n-k;\delta) \ge F_\alpha(k-1, n-k)\}$. It increases in $\delta$: $F_n \overset{d}{=} \frac{(V+Z^2)/(k-1)}{W/(n-k)}$ ($V \sim \chi^2(k-2)$, $Z \sim N(\sqrt{\delta},1)$, $W \sim \chi^2(n-k)$ independent), so the power is $P_\theta(Z^2 \ge Y)$, $Y \perp Z$, $\theta = \sqrt{\delta}$, and

$$\frac{d}{d\theta} P_\theta(Z^2 \ge y) = e^{-y/2}\phi(\theta)\big(e^{\sqrt{y}\theta} - e^{-\sqrt{y}\theta}\big) > 0 \qquad (\theta > 0).$$

Likewise the balanced interaction test has power $P\{F((a-1)(b-1), n-ab; \delta_{AB}) \ge F_\alpha\}$, $\delta_{AB} = \sum r\gamma_{ij}^2/\sigma^2$.

## 10.4 Regression analysis

**Motivation.** Regression is the same linear-model theory with continuous regressors.

The normal linear regression model (Chapters 4, 6):

$$\mathbf{Y} = X\beta + \mathbf{e}, \qquad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \qquad \beta \in \mathbb{R}^{p+1},\ \sigma^2 > 0,\ \operatorname{rank} X = p+1.$$

**Theorem 10.12**

(a) The UMVUEs of $\mathbf{c}'\beta$ and $\sigma^2$ are $\mathbf{c}'\hat\beta = \mathbf{c}'(X'X)^{-1}X'\mathbf{Y}$ and $\hat\sigma^2 = \|\mathbf{Y}-X\hat\beta\|^2/(n-p-1)$.

(b) $\hat\beta \sim N_{p+1}(\beta, \sigma^2(X'X)^{-1})$.

(c) $\hat\beta$ and $\hat\sigma^2$ are independent.

(d) $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$.

**Proof.** The density is a full-rank exponential family, so $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ is complete sufficient; $X'\mathbf{Y} = (X'X)\hat\beta$ and $\mathbf{Y}'\mathbf{Y} = (n-p-1)\hat\sigma^2 + \hat\beta'(X'X)\hat\beta$, so the unbiased $\mathbf{c}'\hat\beta$, $\hat\sigma^2$ (Theorem 6.16) are UMVUEs. (b)–(d): Theorem 4.19 in Volume I. ■

**Theorem 10.13 (Confidence sets for $\beta$)**

For a $(p+1)\times r$ matrix $C$ of rank $r$:

(a) $P\{(C'\beta - C'\hat\beta)'(C'(X'X)^{-1}C)^{-1}(C'\beta - C'\hat\beta) \le r\hat\sigma^2 F_\alpha(r, n-p-1)\} = 1-\alpha$;

(b) $P\Big\{|\mathbf{c}'\beta - \mathbf{c}'\hat\beta| \le \sqrt{\mathbf{c}'(X'X)^{-1}\mathbf{c}\,\hat\sigma^2}\sqrt{rF_\alpha(r, n-p-1)} \;\; \forall \mathbf{c} \in \mathrm{col}(C)\Big\} = 1-\alpha$.

Proof as for Theorem 10.3.

To test a subset of coefficients (e.g. all slopes), partition

$$\mathbf{Y} = X_0\beta_0 + X_1\beta_1 + \mathbf{e}, \qquad \operatorname{rank} X_0 = p_0,\ \operatorname{rank}(X_0, X_1) = p_0 + p_1,$$

with $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$, $X_{1|0} = (I-\Pi_0)X_1$, $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, $\Pi_{0,1} = X(X'X)^{-1}X' = \Pi_0 + \Pi_{1|0}$, $\Pi_0\Pi_{1|0} = 0$ (Theorem 6.17).

**Theorem 10.14 (Testing a subset of coefficients)**

For $H_0 : \beta_1 = \mathbf{0}$ vs $H_1 : \beta_1 \ne \mathbf{0}$:

(a) $\min_{\beta_0}\|\mathbf{Y} - X_0\beta_0\|^2 = \min_{\beta_0,\beta_1}\|\mathbf{Y} - X_0\beta_0 - X_1\beta_1\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$.

(b) With $R(1|0) = \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$ and $SSE = \mathbf{Y}'(I - \Pi_0 - \Pi_{1|0})\mathbf{Y}$, the LRT rejects when

$$F_n = \frac{R(1|0)/p_1}{SSE/(n-p_0-p_1)} \ge F_\alpha(p_1, n-p_0-p_1).$$

(c) $F_n \sim F(p_1, n-p_0-p_1; \delta)$ with $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$, and the power increases in $\delta$.

**Proof.** (a) The minima are $\mathbf{Y}'(I-\Pi_0)\mathbf{Y}$ and $\mathbf{Y}'(I-\Pi_{0,1})\mathbf{Y}$, with $I - \Pi_0 = (I-\Pi_{0,1}) + \Pi_{1|0}$.

(b) The MLEs of $\sigma^2$ are $SSE/n$ and $(SSE + R(1|0))/n$, giving $n\log(1 + R(1|0)/SSE)$.

(c) $\Pi_{1|0}$ is idempotent of trace $p_1$, so $R(1|0)/\sigma^2 \sim \chi^2(p_1;\delta)$, $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$ ($\Pi_{1|0}X_0 = 0$); $SSE/\sigma^2 \sim \chi^2(n-p_0-p_1)$, independent since $\Pi_{1|0}(I-\Pi_0-\Pi_{1|0}) = 0$. Monotone power as in Example 10.2. ■

**Theorem 10.15 (General linear hypotheses)**

For a $(p+1)\times r$ matrix $C$ of rank $r$, the LRT of $H_0 : C'\beta = \mathbf{0}$ vs $H_1 : C'\beta \ne \mathbf{0}$ satisfies:

(a) with $\Pi_{1|0} = X(X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'(X'X)^{-1}X'$,

$$\min_{C'\beta=\mathbf{0}}\|\mathbf{Y} - X\beta\|^2 = \min_{\beta}\|\mathbf{Y} - X\beta\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y},$$

$$\mathbf{Y}'\Pi_{1|0}\mathbf{Y} = (C'\hat\beta)'\{C'(X'X)^{-1}C\}^{-1}(C'\hat\beta);$$

(b) with $SSE = \mathbf{Y}'(I - X(X'X)^{-1}X')\mathbf{Y}$, it rejects when $\dfrac{\mathbf{Y}'\Pi_{1|0}\mathbf{Y}/r}{SSE/(n-p-1)} \geq F_\alpha(r, n-p-1)$;

(c) this statistic is $F(r, n-p-1; \delta)$ with $\delta = \beta' C\{C'(X'X)^{-1}C\}^{-1}C'\beta/\sigma^2$.

**Proof.** Reduce to Theorem 10.14: take orthonormal $C_1$ spanning $\mathrm{col}(C)$, extended to an orthonormal basis $(C_0\ C_1)$, so $C = C_1 B$, $B$ invertible. With $D_0 = XC_0$, $D_1 = XC_1$, $\gamma_0 = C_0'\beta$, $\gamma_1 = C_1'\beta$,

$$X\beta = X(C_0\ C_1)(C_0\ C_1)'\beta = D_0\gamma_0 + D_1\gamma_1,$$

and $C'\beta = \mathbf{0} \iff \gamma_1 = \mathbf{0}$. So $\Pi_{1|0}$ projects onto the complement of $\mathrm{col}(D_0)$ in $\mathrm{col}(X)$:

$$\{X\beta : \beta' X'XC_0 = \mathbf{0}\} = \{X\beta : X'X\beta \in \mathrm{col}(C)\} = \mathrm{col}(X(X'X)^{-1}C).$$

With $S = X(X'X)^{-1}C$, $\Pi_{1|0} = S(S'S)^{-1}S'$, and $S'\mathbf{Y} = C'\hat\beta$. ∎

**Linear models.** ANOVA models are regressions with dependent design columns, identified by constraints:

$$\mathbf{Y} = X\beta + \mathbf{e}, \quad C'\beta = \mathbf{0}, \quad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \quad \operatorname{rank} X = p + 1 - r < p + 1,$$

with the rows of $X$ and $C'$ jointly independent: a **linear model**.

**Example 10.3 (One-way model as a linear model)**

$X_{ij} = \bar\mu + \alpha_i + e_{ij}$, $\sum n_i\alpha_i = 0$: with $\mathbf{Y}$ stacked by groups, $\beta = (\bar\mu, \alpha_1, \ldots, \alpha_k)'$, $\mathbf{c} = (0, n_1, \ldots, n_k)'$,

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{0} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & & & \ddots & \\ \mathbf{1}_{n_k} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{n_k} \end{pmatrix}.$$

The **design matrix** has rank $k < k+1$; $\mathbf{c}'\beta = 0$ identifies $\beta$. With $X'X$ singular, use projections instead of $(X'X)^{-1}$; the theory goes through as in Theorem 10.15.